

Question ID: 25da87f8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

A triangle with angle measures 30° , 60° , and 90° has a perimeter of $18 + 6\sqrt{3}$. What is the length of the longest side of the triangle?

Rationale

The correct answer is 12. It is given that the triangle has angle measures of 30° , 60° , and 90° , and so the triangle is a special right triangle. The side measures of this type of special triangle are in the ratio $2:1:\sqrt{3}$. If x is the measure of the shortest leg, then the measure of the other leg is $\sqrt{3}x$ and the measure of the hypotenuse is $2x$. The perimeter of the triangle is given to be $18 + 6\sqrt{3}$, and so the equation for the perimeter can be written as $2x + x + \sqrt{3}x = 18 + 6\sqrt{3}$. Combining like terms and factoring out a common factor of x on the left-hand side of the equation gives $(3 + \sqrt{3})x = 18 + 6\sqrt{3}$. Rewriting the right-hand side of the equation by factoring out 6 gives $(3 + \sqrt{3})x = 6(3 + \sqrt{3})$. Dividing both sides of the equation by the common factor $(3 + \sqrt{3})$ gives $x = 6$. The longest side of the right triangle, the hypotenuse, has a length of $2x$, or $2(6)$, which is 12.